Lab- Getting Logs from a container

Docker Containers:

```
vijayagagla@DESKTOP-FP4OP26:~/loki-lab-advanced$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS         PORTS                                       NAMES
b0ad18f23b63   grafana/grafana:latest "/run.sh"                18 minutes ago  Up 18 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp grafana
d661a6e42731   busybox                "sh -c 'while true; …"   25 minutes ago  Up 25 minutes                                              sample-app
fb66b4c1d869   grafana/promtail:3.0.0 "/usr/bin/promtail -…"   29 minutes ago  Up 29 minutes                                              promtail
fccb770d950c   grafana/loki:3.0.0     "/usr/bin/loki -conf…"   34 minutes ago  Up 34 minutes  0.0.0.0:3100->3100/tcp, [::]:3100->3100/tcp loki
```

Loki readyness/metrics:

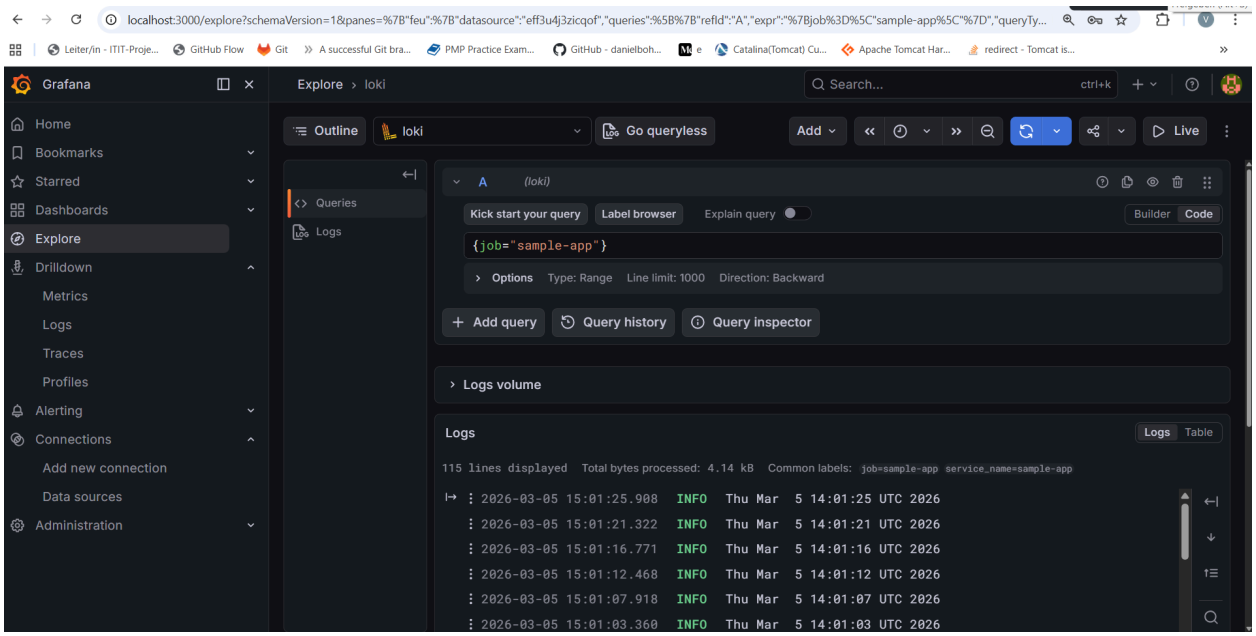
localhost:3100/ready

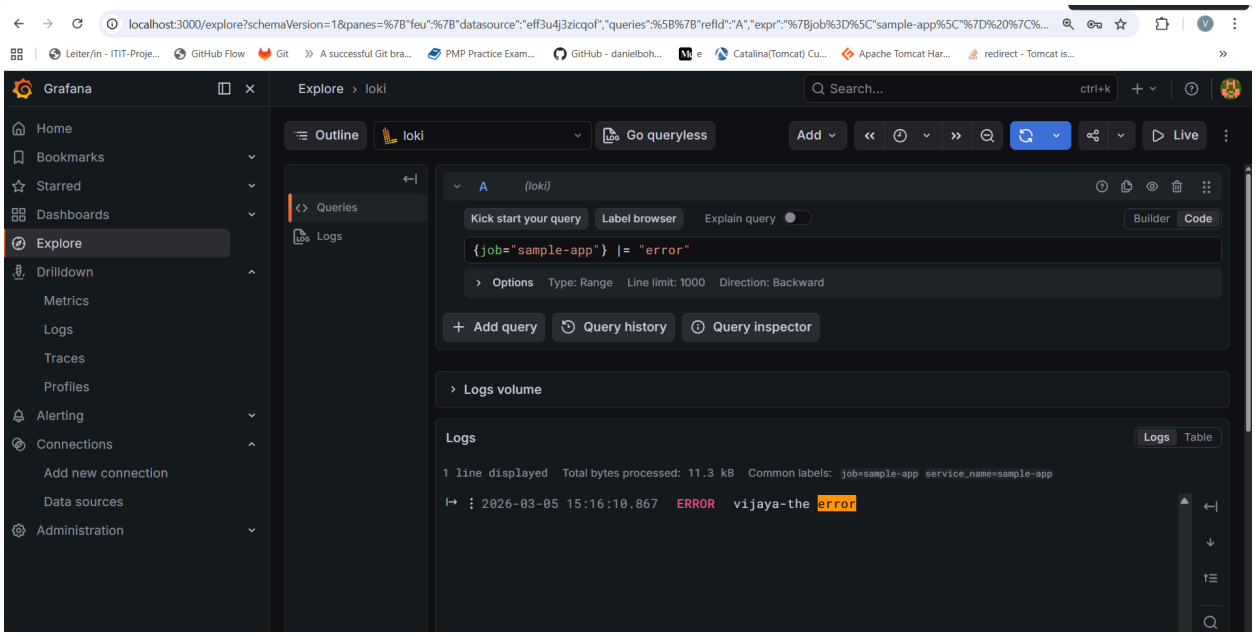
```
ready
```

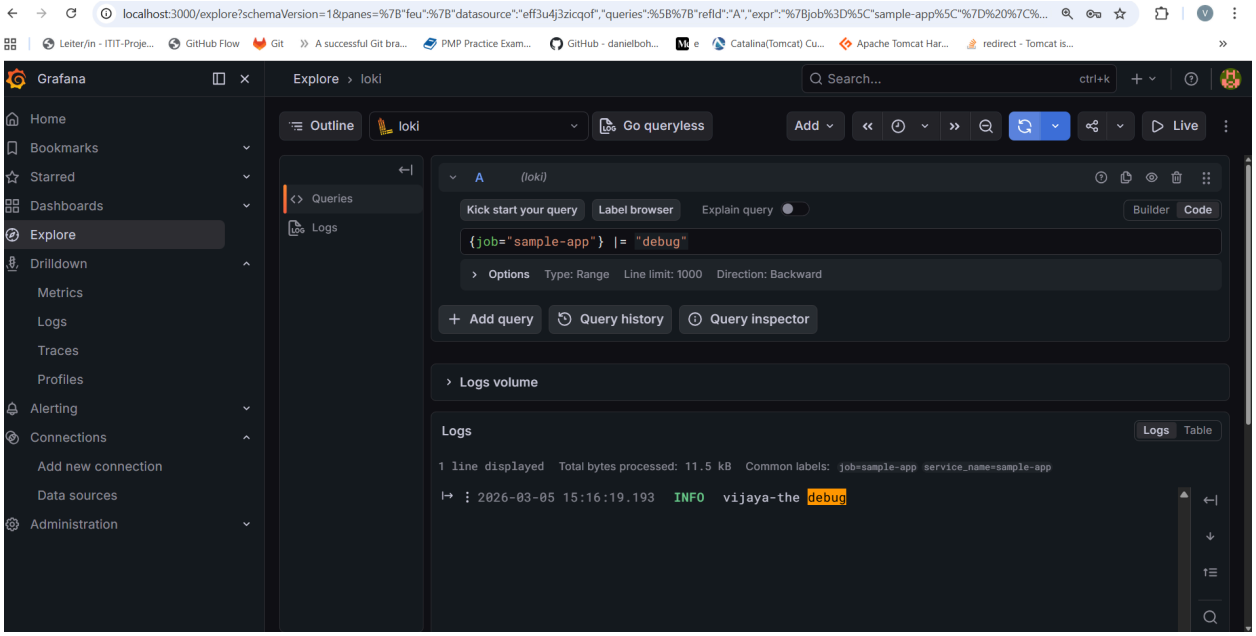
localhost:3100/metrics

```
# HELP deprecated_flags_inuse_total The number of deprecated flags currently set.
# TYPE deprecated_flags_inuse_total counter
deprecated_flags_inuse_total 0
# HELP go_cgo_go_to_c_calls_calls_total Count of calls made from Go to C by the current process.
# TYPE go_cgo_go_to_c_calls_calls_total counter
go_cgo_go_to_c_calls_calls_total 0
# HELP go_cpu_classes_gc_mark_assist_cpu_seconds_total Estimated total CPU time goroutines spent performing GC tasks to assist the GC and prevent it from falling behind the application. This
metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_gc_mark_assist_cpu_seconds_total counter
go_cpu_classes_gc_mark_assist_cpu_seconds_total 0.007789591
# HELP go_cpu_classes_gc_mark_dedicated_cpu_seconds_total Estimated total CPU time spent performing GC tasks on processors (as defined by GOMAXPROCS) dedicated to those tasks. This metric is
an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_gc_mark_dedicated_cpu_seconds_total counter
go_cpu_classes_gc_mark_dedicated_cpu_seconds_total 0.430692503
# HELP go_cpu_classes_gc_mark_idle_cpu_seconds_total Estimated total CPU time spent performing GC tasks on spare CPU resources that the Go scheduler could not otherwise find a use for. This
should be subtracted from the total GC CPU time to obtain a measure of compulsory GC CPU time. This metric is an overestimate, and not directly comparable to system CPU time measurements.
Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_gc_mark_idle_cpu_seconds_total counter
go_cpu_classes_gc_mark_idle_cpu_seconds_total 0.088161151
# HELP go_cpu_classes_gc_pause_cpu_seconds_total Estimated total CPU time spent with the application paused by the GC. Even if only one thread is running during the pause, this is computed as
GOMAXPROCS times the pause latency because nothing else can be executing. This is the exact sum of samples in /gc/pause:seconds if each sample is multiplied by GOMAXPROCS at the time it is
taken. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_gc_pause_cpu_seconds_total counter
go_cpu_classes_gc_pause_cpu_seconds_total 0.137146872
# HELP go_cpu_classes_gc_total_cpu_seconds_total Estimated total CPU time spent performing GC tasks. This metric is an overestimate, and not directly comparable to system CPU time
measurements. Compare only with other /cpu/classes metrics. Sum of all metrics in /cpu/classes/gc.
# TYPE go_cpu_classes_gc_total_cpu_seconds_total counter
go_cpu_classes_gc_total_cpu_seconds_total 0.663790117
# HELP go_cpu_classes_idle_cpu_seconds_total Estimated total available CPU time not spent executing any Go or Go runtime code. In other words, the part of /cpu/classes/total:cpu-seconds that
was unused. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_idle_cpu_seconds_total counter
go_cpu_classes_idle_cpu_seconds_total 27687.278378183
# HELP go_cpu_classes_scavenge_assist_cpu_seconds_total Estimated total CPU time spent returning unused memory to the underlying platform in response eagerly in response to memory pressure.
This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_scavenge_assist_cpu_seconds_total counter
go_cpu_classes_scavenge_assist_cpu_seconds_total 4.84e-07
# HELP go_cpu_classes_scavenge_background_cpu_seconds_total Estimated total CPU time spent performing background tasks to return unused memory to the underlying platform. This metric is an
overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics.
# TYPE go_cpu_classes_scavenge_background_cpu_seconds_total counter
go_cpu_classes_scavenge_background_cpu_seconds_total 0.007996788
# HELP go_cpu_classes_scavenge_total_cpu_seconds_total Estimated total CPU time spent performing tasks that return unused memory to the underlying platform. This metric is an overestimate,
and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes metrics. Sum of all metrics in /cpu/classes/scavenge.
```

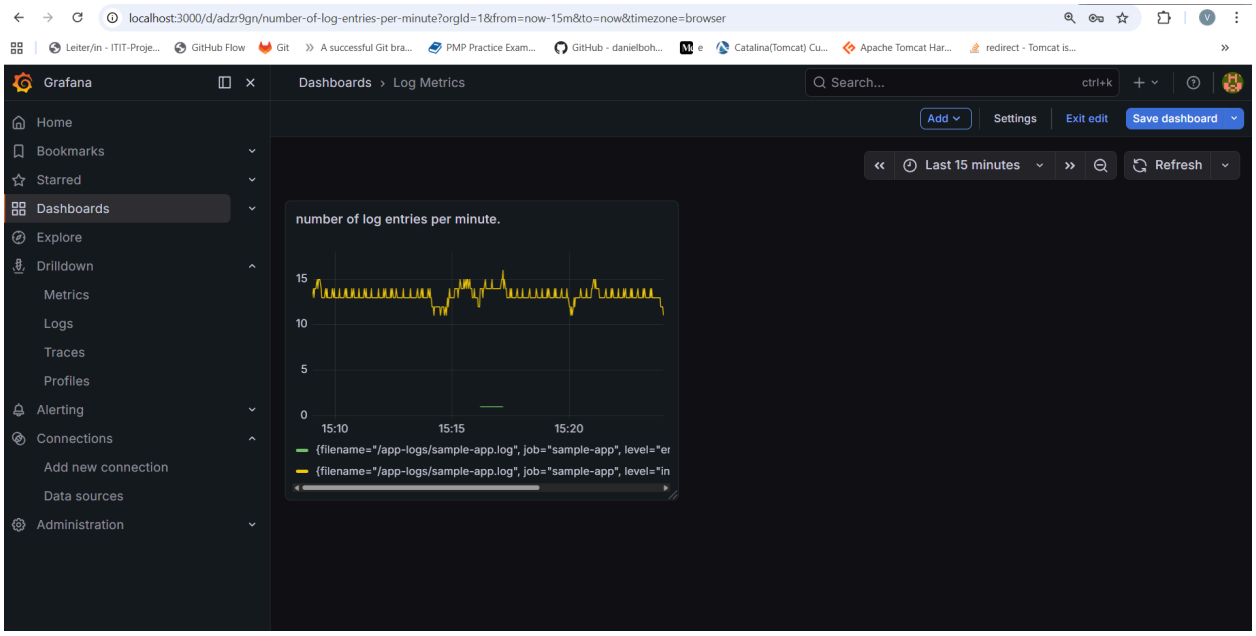
Logs:


localhost:3000/explore?schemaVersion=1&panes=%7B"feu":%7B"datasource":"eff3u4j3zicqof","queries":%5B%7B"refId":"A","expr":"%7Bjob%3D%5C"sample-app%5C"%7D","queryTy...
Grafana
Explore › loki
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Connections
Add new connection
Data sources
Administration
Outline
loki
Go queryless
Add
Live
Queries
Logs
A
(loki)
Kick start your query
Label browser
Explain query
Builder
Code
{job="sample-app"}
Options
Type: Range
Line limit: 1000
Direction: Backward
Add query
Query history
Query inspector
Logs volume
Logs
Logs
Table
115 lines displayed
Total bytes processed: 4.14 kB
Common labels:
job=sample-app
service_name=sample-app
2026-03-05 15:01:25.908 INFO Thu Mar 5 14:01:25 UTC 2026
2026-03-05 15:01:21.322 INFO Thu Mar 5 14:01:21 UTC 2026
2026-03-05 15:01:16.771 INFO Thu Mar 5 14:01:16 UTC 2026
2026-03-05 15:01:12.468 INFO Thu Mar 5 14:01:12 UTC 2026
2026-03-05 15:01:07.918 INFO Thu Mar 5 14:01:07 UTC 2026
2026-03-05 15:01:03.360 INFO Thu Mar 5 14:01:03 UTC 2026

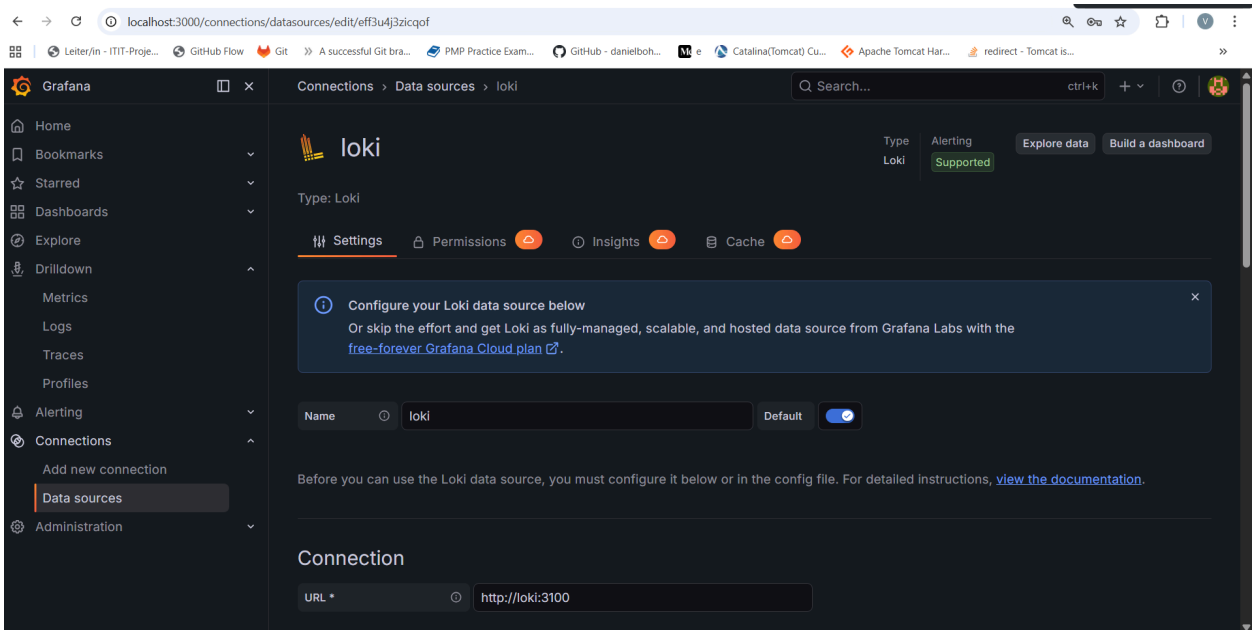

localhost:3000/explore?schemaVersion=1&panes=%7B"feu":%7B"datasource":"eff3u4j3zicqof","queries":%5B%7B"refId":"A","expr":"%7Bjob%3D%5C"sample-app%5C"%7D%20%7C%...
Grafana
Explore › loki
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Connections
Add new connection
Data sources
Administration
Outline
loki
Go queryless
Add
Live
Queries
Logs
A
(loki)
Kick start your query
Label browser
Explain query
Builder
Code
{job="sample-app"} |= "error"
Options
Type: Range
Line limit: 1000
Direction: Backward
Add query
Query history
Query inspector
Logs volume
Logs
Logs
Table
1 line displayed
Total bytes processed: 11.3 kB
Common labels:
job=sample-app
service_name=sample-app
2026-03-05 15:16:10.867 ERROR vijaya-the error



Log Metrics Dashboard



Loki as Datasource :


localhost:3000/connections/datasources/edit/eff3u4j3zicqof
Grafana
Connections › Data sources › loki
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Connections
Add new connection
Data sources
Administration
loki
Type
Loki
Alerting
Supported
Explore data
Build a dashboard
Type: Loki
Settings
Permissions
Insights
Cache
Configure your Loki data source below
Or skip the effort and get Loki as fully-managed, scalable, and hosted data source from Grafana Labs with the free-forever Grafana Cloud plan.
Name
loki
Default
Before you can use the Loki data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Connection
URL *
http://loki:3100